



Report 1

Anatomy of a Web Request

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**NETWORK SECURITY –
BATCH 2**



Introduction:

Whenever a user enters a website such as <https://www.google.com> in Google Chrome, the browser performs many operations before the webpage appears. These operations include URL parsing, browser cache lookup, DNS resolution, ARP discovery, TCP connection establishment, TLS handshake, HTTP request generation, packet routing, server processing, response delivery, browser rendering and page display. Although the user only sees webpage loading, hundreds of packets travel through multiple networking devices. The OSI Reference Model provides a structured way to understand this complete communication process.

Objectives:

Understand how Chrome retrieves a webpage. Map every operation to the seven OSI layers. Explain protocols used at each layer. Identify real-world cyber threats. Recommend practical security controls. Understand how secure web communication works.

Scenario:

A user enters www.google.com in the Chrome browser and presses Enter the request travels through all OSI layers to retrieve the webpage from the web server and display it on the screen

1. Application Layer (Layer 7):

- User enters a URL in Chrome.
- Chrome checks browser cache for the webpage.
- If not found, Chrome sends a **DNS request** to resolve the domain name (www.google.com) into an IP address.
- Chrome creates an HTTPS request to access the website.

Protocols:

- HTTP/HTTPS
- DNS

Threats:

- DNS Spoofing
- Phishing Websites

- Cross-Site Scripting (XSS)
- SQL Injection
- Malware Downloads

Mitigation

- Use HTTPS websites
- Enable DNSSEC
- Use Web Application Firewalls (WAF).
- Browser security updates.
- Anti-malware software

2. Presentation Layer (Layer 6):

- Data is translated into a format understood by both systems.
- HTTPS uses SSL/TLS encryption to secure communication.
- Certificates are verified.
- Data compression may occur to improve speed.

Protocols:

- SSL/TLS
- JPEG, PNG, GIF
- ASCII, Unicode

Threats:

- Man-in-the-Middle (MITM) Attack
- Weak Encryption
- Certificate Forgery

Mitigation:

- Strong TLS versions (TLS 1.3)
- Valid Digital Certificates
- Certificate Pinning
- Strong Encryption Algorithms

3. Session Layer (Layer 5):

- Establishes and manages the communication session between browser and server.
- Maintains HTTPS session.
- Handles authentication and session IDs.
- Keeps the connection active until webpage transfer completes.

Threats:

- Session Hijacking
- Session Replay Attacks
- Session Fixation

Mitigation:

- Secure Session Tokens
- Multi-Factor Authentication (MFA)
- Session Timeout
- Secure Cookies

4. Transport Layer (Layer 4):

- TCP connection is established using the Three-Way Handshake.
 1. Client → SYN
 2. Server → SYN-ACK
 3. Client → ACK
- Data is divided into segments.
- Reliability, error checking, and flow control are provided.
- HTTPS typically uses TCP Port 443.

Protocols:

- TCP
- UDP (for DNS queries)

Threats:

- SYN Flood Attack
- Port Scanning
- TCP Session Hijacking

Mitigation:

- Firewalls
- Intrusion Prevention Systems (IPS)
- SYN Cookies
- Rate Limiting

5. Network Layer (Layer 3):

- Source and destination IP addresses are added.
- Router determines the best path to the destination server.
- Packets travel through multiple routers across the Internet.

Protocols

- IP
- ICMP
- OSPF
- BGP

Threats:

- IP Spoofing
- Routing Attacks
- DDoS Attacks

Mitigation:

- Access Control Lists (ACLs)
- Anti-Spoofing Filters
- VPNs
- DDoS Protection Systems

6. Data Link Layer (Layer 2):

- IP packet is encapsulated into a frame.
- Device uses **ARP** to find the MAC address of the default gateway.
- Frames are transmitted over the local network.

Protocols:

- Ethernet
- Wi-Fi (802.11)
- ARP

Threats:

- ARP Spoofing
- MAC Flooding
- VLAN Hopping

Mitigation:

- Dynamic ARP Inspection (DAI)
- Port Security
- VLAN Segmentation
- Switch Security Features

7. Physical Layer (Layer 1):

- Data is converted into electrical signals, radio waves, or light signals.
- Signals travel through:
 - Ethernet cables
 - Fiber optic cables
 - Wi-Fi signals

Threats:

- Cable Tapping
- Signal Interception
- Physical Device Theft
- Hardware Tampering

Mitigation:

- Physical Security Controls
- Secure Data Centers
- Fiber Optic Monitoring
- Network Access Control (NAC)

Encapsulation Process:

When sending data:

Application Data

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Presentation Layer Encryption

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Session Management Information

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TCP Segment

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IP Packet

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Ethernet/Wi-Fi Frame

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Physical Signals

Decapsulation Process:

At the Web Server:

Physical Signals

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Frame

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Packet

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TCP Segment

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Session Information

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Decryption (TLS)

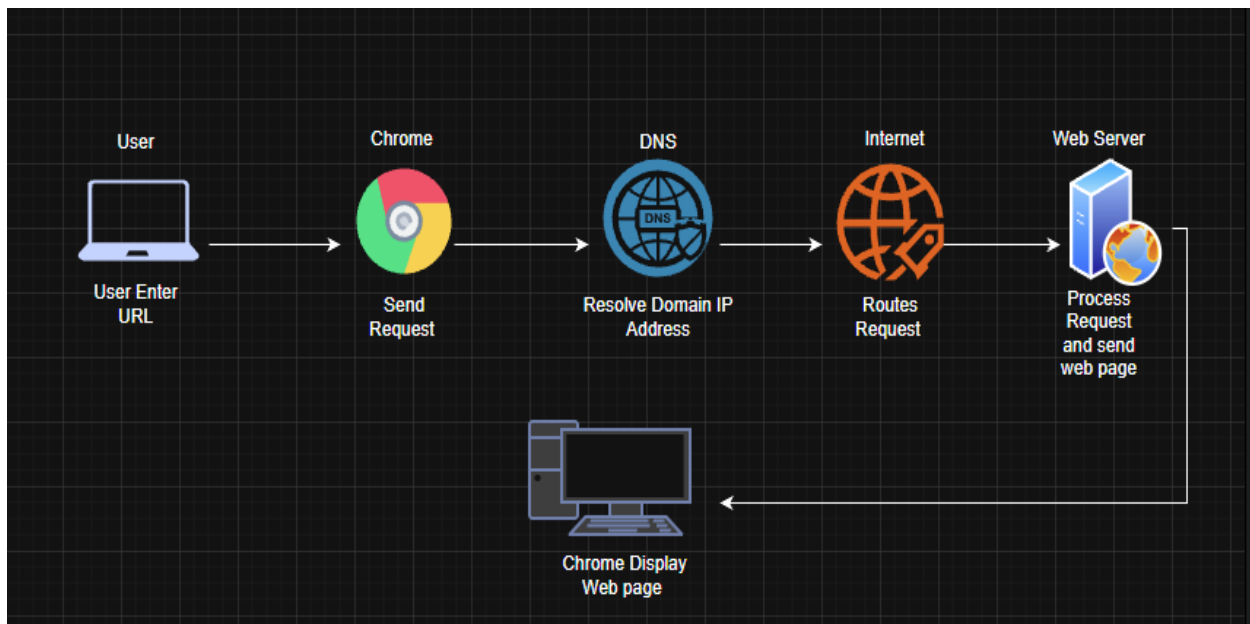
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HTTP Request

Complete Workflow Summary:

1. User enters www.google.com in Chrome.
2. DNS resolves the domain to an IP address.
3. TLS handshake establishes a secure HTTPS connection.
4. TCP three-way handshake creates a reliable connection.
5. IP packets are routed across the Internet.
6. Ethernet/Wi-Fi frames carry packets through local networks.
7. Physical media transmits signals.
8. The web server receives the request and sends the webpage.
9. Chrome receives the response, decrypts it, render HTML/CSS/JavaScript, and displays the webpage to the user.

Diagram: (Draw.io)



Conclusion:

The OSI model provides a structured process for webpage delivery. Each layer performs specific functions from user interaction at the Application Layer to signal transmission at the Physical Layer. Security threats exist at every layer, but mechanisms such as TLS encryption, firewalls, IPS, secure sessions, VLAN security, and physical protections help ensure confidentiality, integrity, and availability of web communication.